TrES-2b

R-0099

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_180413` · created 2026-07-16T02:56:07+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458221.91508 +/- 0.00057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.022
Transit depth (Rp/Rs)^2	2.15 +/- 0.65 [%]
Orbital Inclination (inc)	83.77 +/- 1.78
Airmass coefficient 1 (a1)	0.986 +/- 0.012
Airmass coefficient 2 (a2)	0.011 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.2 %
Best Comparison Star	#1 - [108, 129]
Optimal Aperture	4.14
Optimal Annulus	15.11
Transit Duration (day)	0.066 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.147 ± 0.022 vs 0.1254 (deviation 0.98σ)
Duration fit vs published	0.066 d vs 0.0746 d (deviation 0.26σ)
Mid-time O–C	1.3 min
Depth SNR	6.7
Residual scatter	1.2 %

Light curve

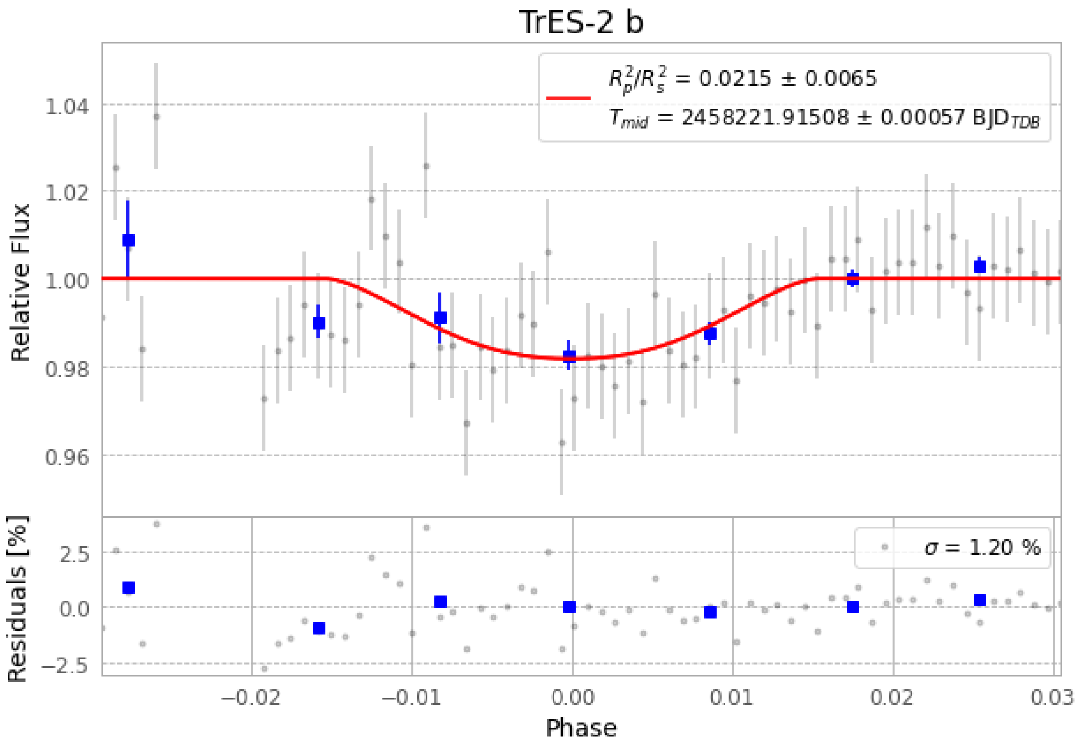

Figure 1 — FinalLightCurve_TrES-2 b_2018-04-13.png (SHA-256 99adc6b45a2e3c28…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [432, 210], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 40007b353b791906…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

72 FITS frames (47 MB), `TRES-2180413081212.FITS` → `TRES-2180413114512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-180413.log (fd203d22c478…), FinalLightCurve_TrES-2 b_2018-04-13.png (99adc6b45a2e…), FinalLightCurve_TrES-2 b_2018-04-13.csv (47ba60c7e051…)

Compute resources

Wall time 305.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)